Lec 16

Recap

0. Total Law of Probability
1. Conditional Expectation

$$E[X \mid Y=y] = \sum_{x \in \mathcal{X}_0} P_{X|Y}(x|y)\, x$$

$$E[X \mid A] = \sum_{x \in \mathcal{X}_0} P_{X|A}(x)\, x$$

Let $f(y) = E[X \mid Y=y]$

then define

$E[X \mid Y] = f(Y)$

2. Total Law of Expectation

ⓐ $A_1, \ldots A_n$ partition Ω.

$$E[X] = \sum_{i=1}^{n} P(A_i)\, E[X \mid A_i]$$

ⓑ $E[X] = E\left[\, E[X \mid Y] \,\right]$, $E[Y] = E\left[E[Y \mid X] \right]$

$E[g(X)] = E\left[E[g(X) \mid Y] \right]$.

3. Conditional Variance

$$\text{Var}(X \mid Y=y) = E\left[(X - E[X \mid Y=y])^2 \mid Y=y \right]$$

Show that

$\text{Var}(X \mid Y=y) = E[X^2 \mid Y=y] - (E[X \mid Y=y])^2$

Excercise

$\boxed{\text{Var}(X \mid Y)}$

R.V that is a function of Y.

$$X = \begin{cases} 1 & \text{w.p } p \\ 2 & \text{w.p } 1-p \end{cases}$$

$Y | X=1 \sim \text{Poisson}(\lambda_1)$

$Y | X=2 \sim \text{Poisson}(\lambda_2)$

$Z \sim \text{Poisson}(\lambda)$

$Var(Z) = \lambda$

$E[Z] = \lambda$

$\Rightarrow E[Z^2] = Var(Z) + (E[Z])^2$

$= \lambda + \lambda^2$

$$E[Y] = E[E[Y|X]]$$

$$= P_X(1)\, E[Y|X=1] + P_X(2)\, E[Y|X=2] \rightarrow (1)$$

$$E[Y] = p\lambda_1 + (1-p)\lambda_2$$

$$\boxed{Var(Y) = E[Y^2] - (E[Y])^2}$$

$$E[Y^2] = pE[Y^2|X=1] + (1-p)E[Y^2|X=2] \rightarrow (2)$$

$$= p[\lambda_1 + \lambda_1^2] + (1-p)[\lambda_2 + \lambda_2^2]$$

$Var(Y)$

$E[$

$Var($

$=$

(-1)

$\mathrm{Var}(Z) + (E[Z])^2$

$+\lambda^2$

$\rightarrow ①$

$\rightarrow ②$

$$\mathrm{Var}(Y) = p(\lambda_1 + \lambda_1^2) + (1-p)(\lambda_2 + \lambda_2^2)$$
$$- \left[p\lambda_1 + (1-p)\lambda_2 \right]^2$$

$$E\left[\mathrm{Var}(Y|X) \right] \neq \mathrm{Var}(Y)$$

$$\mathrm{Var}(Y|X=1)\,P(X=1) + P(X=2)\,\mathrm{Var}(Y|X=2)$$

$$= \lambda_1 p + (1-p)\lambda_2$$

Total Law of Variance

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

$$E[Var(X|Y)] = \sum_{y \in \mathcal{Y}} P_Y(y)\, Var(X|Y=y)$$

$$= \sum_{y \in \mathcal{Y}} P_Y(y) \left[E[X^2|Y=y] - \left(E[X|Y=y]\right)^2 \right] \quad \text{(from Excercise problem)}$$

$$= \sum_{y \in \mathcal{Y}} P_Y(y) E[X^2|Y=y] - \sum_{y \in \mathcal{Y}} P_Y(y) \left(E[X|Y=y]\right)^2$$

$$= E\left[E[X^2|Y]\right] - E\left[\left(E[X|Y]\right)^2\right]$$

$$= E[X^2] - E\left[\left(E[X|Y]\right)^2\right]$$

$$\mathrm{Var}(\underbrace{E[X|Y]}_{:=f(Y)}) = E\left[\left(f(Y) - E[f(Y)]\right)^2\right]$$

$$E[f(Y)] = E[E[X|Y]]$$
$$= E[X]$$

$$\mathrm{Var}(f(Y)) = E\left[(f(Y))^2\right] - \left(E[f(Y)]\right)^2$$
$$= E\left[(f(Y))^2\right] - (E[X])^2 \longrightarrow 4$$

3 + 4

$$\mathrm{Var}\left(E[X|Y]\right) + E\left[\mathrm{Var}(X|Y)\right] = E[X^2] - \left(E[X]\right)^2$$
$$= \mathrm{Var}(X).$$

Example:

$$\mathrm{Var}(Y) = \mathrm{Var}(E[Y|X]) + E[\mathrm{Var}(Y|X)]$$

$$E[\mathrm{Var}(Y|X)] = p\lambda_1 + (1-p)\lambda_2$$

$$\mathrm{Var}(\underbrace{E[Y|X]}_{\hat{f}(X)}) = E[(\hat{f}(X))^2] - (E[\hat{f}(X)])^2$$

$$\hat{f}(x) = E[Y|X=x]$$

$$= \begin{cases} \lambda_1 & x=1 \\ \lambda_2 & x=2 \end{cases}$$

$$E[\hat{f}(X)^2] = P_X(1)\,\hat{f}(1)^2 + P_X(2)\,\hat{f}(2)^2$$

$$= p\lambda_1^2 + (1-p)\lambda_2^2$$

$$\mathrm{Var}(Y) = p(\lambda_1+\lambda_1^2)+(1-p)(\lambda_2+\lambda_2^2) - \left[p\lambda_1+(1-p)\lambda_2\right]^2$$

$$E[\hat{f}(x)] = E\left[E[Y|X]\right] = E[Y] = p\lambda_1+(1-p)\lambda_2$$

$$\mathrm{Var}(\hat{f}(x)) = E[\hat{f}(x)^2] - \left(E[\hat{f}(x)]\right)^2$$

$$= \left(p\lambda_1^2+(1-p)\lambda_2^2\right) - \left(p\lambda_1+(1-p)\lambda_2\right)^2$$

$$E\left[\mathrm{Var}(Y|X)\right] + \mathrm{Var}\left(\underbrace{E[Y|X]}_{\hat{f}(x)}\right) = p\lambda_1+(1-p)\lambda_2+p\lambda_1^2+(1-p)\lambda_2^2 - \left(p\lambda_1+(1-p)\lambda_2\right)^2$$

Independence of two R.Vs

R.Vs X and Y are said to be independent if

$$P_{X,Y}(x,y) = P_X(x)\,P_Y(y) \quad \forall x \in \mathcal{X}, \forall y \in \mathcal{Y}.$$

$$\Longleftrightarrow \quad P_{X|Y}(x|y) = P_X(x) \quad \forall x \in \mathcal{X}, y \in \mathcal{Y}$$

$$\Longleftrightarrow \quad P_{Y|X}(y|x) = P_Y(y) \quad \forall x \in \mathcal{X}, \forall y \in \mathcal{Y}.$$

RVs $X_1, X_2, \ldots X_n$ are said to be independent

$$P_{X_1, X_2, \ldots X_n}(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P_{X_i}(x_i) \quad \forall (x_1, x_2, \ldots x_n) \in \mathcal{X}_1 \times \mathcal{X}_2 \times \ldots \mathcal{X}_n$$

Independence of $X_1, \ldots X_n$ implies

(a) pairwise independence of any two R.Vs among $X_1, \ldots X_n$

(b) any subset of R.Vs from $X_1, \ldots X_n$ are independent.

X, Y independent implies

(a) $E[XY] = E[X]\,E[Y]$

(b) $E[h(X)\,g(Y)] = E[h(X)]\,E[g(Y)]$

(c) $Var(X+Y) = Var(X) + Var(Y)$

For any R.Vs X, Y

$$Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$$

where

$$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$$
$$= E[XY] - E[X]E[Y]$$

$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i)$ if $X_1, X_2, \ldots X_n$ are independant.